

TIANER ZHOU

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🎓 EDUCATION

University of Michigan, Ann Arbor, USA

2019 – 2022

Bachelor of Science in Computer Science (Summa Cum Laude)

- **GPA:** 3.917/4.0
- **Selected Coursework:** Machine Learning, Distributed Systems, Operating Systems, Computer Architecture, Linear Algebra, Database, Computer Security, Web Systems.

🔧 AI SYSTEMS & RESEARCH PROJECTS

- **JEPA World Model Research:** Reproduced hierarchical JEPA from HWM_PLDM and optimized its planning / actor loop for JEPA-style world models, achieving up to 10x planning speedup while continuing accuracy-improvement experiments.
- **Latent Ensemble World Models:** Trained 3-member leWM ensembles for model-based planning, using ensemble disagreement as an uncertainty signal to guide actor optimization and trajectory scoring.
- **GPT Training / Inference Stack:** Built GPT-style training/inference from scratch: BPE tokenization, Transformer blocks, RoPE, FlashAttention kernel, and distributed KV cache. [\[repo link\]](#)
- **Offline RL Framework:** Implemented SAC, IQL, AWAC, and PPO; IQL reached 230/300 on Ant Maze in 30k steps. [\[repo link\]](#)
- **AI Runtime Contributions:** Merged PRs to NVIDIA Dynamo and vLLM, including KV-router scheduler tests and an out-of-bounds fix; submitted Ironclaw setup/safety fixes and issues. [\[dynamo\]](#) [\[vLLM\]](#) [\[ironclaw\]](#)
- **LLM Agent Tooling:** Developed a HarmonyOS MCP server enabling LLM agents to inspect and control HarmonyOS device UI. [\[repo link\]](#)

👥 WORK EXPERIENCE

Huawei Technologies Co., Ltd., Hangzhou

Aug 2022 – Present

Graphics Engineer HarmonyOS Graphics Foundation, now deployed on 50M+ devices.

- **Rendering Pipeline & Shader Optimization:** Developed production 2D visual-effect shaders and an asynchronous color-extraction algorithm for adaptive glass effects; built automated parameter-search tooling to tune shader parameters and improve runtime performance up to 10%.
- **Model Fine-tuning and Deployment:** Quantized 3B-parameter image-generation models for resource-constrained mobile deployment, generating 512x512 images on Mate 80 Pro in under 5 seconds. Curated a style-transfer dataset and fine-tuned the model with LoRA for photorealistic-to-stylized image generation.

UI Framework Engineer native UI framework of HarmonyOS, supporting the development of 20,000+ apps.

- **Open Source Contribution:** Core contributor to the UI engine repository (C++/TypeScript), ranked Top 1 with 1200+ commits among 1300+ contributors. [\[old repo link\]](#)
- **Performance Optimization:** Built lazy/reusable scroll containers cutting jank by 60%; designed deduplicated image caching; created a ReactNative-SVG renderer 20% faster than the Android solution.

⚙️ TECHNICAL SKILLS

AI Systems: World Models, JEPA, Model-based Planning, RL, LLM Training/Inference, KV Cache, Attention Kernels · **Programming Languages:** C++, TypeScript, Rust, Go, Python · **Systems:** Distributed Systems, Graphics Runtime, Compiler, xv6 Kernel · **AI Harness:** Codex, Claude Code, Openclaw